

Test Statistical Methods Final 2025



Test ID: 299582 (In progress)

Folder: /Top/SIS/X_401020

Version: 2.0

Randomised: No

Last modified: Wednesday, 22 october 2025 17:18:00

Number of questions: 16

Question order: Random

Display questions once: No

Tools: Calculator extended, Digital scratch paper

Test time: -

Cut-off: Split linear: 50.00 % (Exclude chance score)

Maximum score: 16 pt.

Chance score: 1.97 pt. / 12%

Mark scale: One decimal

Grading series: 1 - 10

Scoring rule: No point deduction for incorrect answer

In test set with: -

Test instruction

Welcome to the exam of the course **Statistical Methods** (X_401020).

The following tools are permitted:

- calculator (non-programmable)
- [formula sheet](#) from Canvas

Important: When you are asked to fill in a numerical value, note that a **decimal point** is used. If you use a comma instead, your answer will be considered incorrect.

For questions about the content of the test you can contact the teacher. Information about the review of the exam is provided via Canvas.

If you have **not** signed up for this exam, you will not receive a result. Through VUnet you can object to the fact that you can no longer sign up after the expiry of the registration deadline (and the fact that you will not receive a result for this exam). Submit your appeal online within one week after the exam. More information can be found at www.vu.nl/intekenen.

Welkom bij de toets van het vak **Statistical Methods** (X_401020).

Bij deze toets mag je gebruik maken van de volgende hulpmiddelen:

- rekenmachine (niet programmeerbaar)
- [formuleblad](#) van Canvas

Belangrijk: Wanneer u een numerieke waarde moet invullen, houd er dan rekening mee dat er een **decimale punt** wordt gebruikt. Als u in plaats daarvan een komma gebruikt, wordt uw antwoord als onjuist beschouwd.

Voor vragen over de inhoud van de toets kun je bij de docent terecht. Informatie over de toetsinzage wordt via Canvas verstrekt.

Als je **niet** hebt ingetekend voor dit tentamen, ontvang je geen resultaat. Via VUnet kun je bezwaar maken tegen het feit dat je niet meer kunt intekenen na het verstrijken van de intekendeadline (en je dus geen resultaat krijgt voor dit tentamen). Dien je bezwaarschrift online in binnen één week na het tentamen. Meer informatie vind je op www.vu.nl/intekenen.

Question 1 – confidence interval – 508892.2.2

Question type: Multiple choice

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Final 2025

Answer option order: Random

Partial scoring: No

Maximum score: 1

Chance score: 0.25 pt. / 25%

Status:

Last modified: 22/10/2025 17:25

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Let $X \sim N(\theta, 1)$ be a normally distributed random variable with unknown expected value but known variance. Running the underlying random experiment 4 times, assume that $x_1=2, x_2=3, x_3=1, x_4=2$ is a realisation of the corresponding random sample $X_1, X_2, X_3, X_4 \sim X$ of size 4. Which of the following intervals is (approximately) the resulting 90% confidence interval for the expected value θ (rounded to 2 decimals)?

A [1.18, 2.82]

B [1.36, 2.64]

C [1.22, 2.78]

D [1.11, 2.11]

Feedback at question level

$[2 - z_{0.05} / 2, 2 + z_{0.05} / 2]$

Question 2 – EX discrete – 508696.4.0

Question type: Multiple response

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Final 2025

Answer option order: Random

Partial scoring: Yes

Maximum score: 1

Chance score: 0.33 pt. / 33%

Status:

Last modified: 06/10/2025 19:50

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Which of the following random variables X has expected value $EX = 2$? Give all correct answers.

A $X \sim \text{Geometric}(0.5)$

B $X \sim \text{Binomial}(40, 0.05)$

C $X \sim \text{Uniform}(0, 2)$

D $X \sim \text{Exponential}(2)$

Question 3 – EX–quadrat – 508706.1.2

Question type: Multiple choice

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Final 2025

Answer option order: Random

Partial scoring: No

Maximum score: 1

Chance score: 0.20 pt. / 20%

Status:

Last modified: 06/10/2025 12:17

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

For a random variable X , let $f_X(x)$ and $F_X(x)$ denote its probability density function and its cumulative distribution function, respectively. Which formula computes the expected value of X^2 ?

- A** $\int x^2 f_X(x) dx$
- B** $\int x^2 F_X(x) dx$
- C** $(\int x f_X(x) dx)^2$
- D** $(\int x F_X(x) dx)^2$
- E** none of the above

Question 4 – hypothesis test – 508896.1.1

Question type: Multiple choice

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Final 2025

Answer option order: Random

Partial scoring: No

Maximum score: 1

Chance score: 0.20 pt. / 20%

Status:

Last modified: 22/10/2025 17:28

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

The speed μ of a car is measured (in km/h). Let $H_0 = \{\mu \leq 50\}$ and $H_1 = \{\mu > 50\}$ denote the null and the alternative hypothesis, respectively; and the rejection region is $K = [55, \infty)$. Assume that the measured speed is $T=54$, while the actual speed is $\mu=52$. After applying the decision rule, what is true?

- A** One makes a type II error.
- B** One makes a type I error.
- C** One makes a type III error.
- D** One makes the correct decision.
- E** No decision is possible.

Feedback at question level

H_0 is accepted, although H_1 is true.

Question 5 – maximum likelihood – 508859.1.2

Question type: Fill in (numerical)

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Final 2025

Answer option order: Fixed

Partial scoring: No

Maximum score: 1

Chance score: 0.00 pt. / 0%

Status:

Last modified: 22/10/2025 17:33

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

A box contains blue and red balls, but the probability θ of drawing a red ball is unknown. Two times you draw balls with replacement until you draw the first red ball. Assume that the realisation x_1, x_2 of the corresponding random sample $X_1, X_2 \sim \text{Geometric}(\theta)$ is $x_1=3, x_2=2$. Estimate θ using the maximum likelihood principle. Give one decimal.

0.4

Feedback at question level

$$L(\theta) = P(X_1 = 3) * P(X_2 = 2) = (1-\theta)^3 \theta^2$$

Question 6 – normal-normal – 508709.1.2

Question type: Multiple choice

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Final 2025

Answer option order: Random

Partial scoring: No

Maximum score: 1

Chance score: 0.20 pt. / 20%

Status:

Last modified: 22/10/2025 17:36

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Assume $X \sim N(2,1)$ and define $Y=2X-1$. Then $Y \sim N(\mu, \sigma^2)$ with

- ☒ **A** $\mu=3, \sigma=2$
- ☐ **B** $\mu=3, \sigma=1$
- ☐ **C** $\mu=2, \sigma=2$
- ☐ **D** $\mu=2, \sigma=1$
- ☐ **E** none of them

Feedback at question level

$$E(Y) = 2E(X)-1, \text{Var}(Y) = 2^2 \text{Var}(X)$$

Question 7 – Pascal – 505727.1.2

Question type: Fill in (numerical)

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Final 2025

Answer option order: Fixed

Partial scoring: No

Maximum score: 1

Chance score: 0.00 pt. / 0%

Status:

Last modified: 22/10/2025 17:39

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

You are waiting in line to pay your bill and there are 5 other people ahead of you. Let us assume that the probability that a random person is paying with cash is 30%. With which probability is the last visitor in front of you the second one paying with cash? Round to two decimals.

0.12

Feedback at question level

$X \sim \text{Pascal}(5, 0.3)$, $P_X(2) = 4 * 0.3^2 * 0.7^3$

Question 8 – PDF => CDF – 508705.1.1

Question type: Fill in (numerical)

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Final 2025

Answer option order: Fixed

Partial scoring: No

Maximum score: 1

Chance score: 0.00 pt. / 0%

Status:

Last modified: 22/10/2025 17:43

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Assume that a random variable X has a probability density function given by $f_X(x) = x^{-2}$ for $x \geq 1$ and $f_X(x) = 0$ for $x < 1$. What is the value of the cumulative distribution function $F_X(x)$ at $x=4$? Write your answer with two decimals.

0.75

Feedback at question level

$F_X(4) = \int_1^4 x^{-2} dx = -1/4 + 1$

Question 9 – PMF => CDF – 508692.2.1

Question type: Fill in (numerical)

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Final 2025

Answer option order: Fixed

Partial scoring: No

Maximum score: 1

Chance score: 0.00 pt. / 0%

Status:

Last modified: 22/10/2025 17:46

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

For the probability mass function of a random variable X we know that $P_X(k) = 2^{-k}$ for all natural numbers $k = 1, 2, \dots$. What is the value of the corresponding cumulative distribution function $F_X(x)$ for $x = 2.5$? Round to two decimals.

0.75

Feedback at question level

$F_X(2.5) = P_X(1) + P_X(2)$

Question 10 – probability normal – 508708.1.1

Question type: Multiple choice

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Final 2025

Answer option order: Random

Partial scoring: No

Maximum score: 1

Chance score: 0.20 pt. / 20%

Status:

Last modified: 22/10/2025 17:20

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Assume $X \sim N(\mu, \sigma^2)$ with $\mu=4$ and $\sigma=2$. Which formula computes $P(X > 2)$?

- ☒ **A** $1 - \Phi(-1)$
- ☐ **B** $1 - \Phi(2)$
- ☐ **C** $\Phi(1)$
- ☐ **D** $\Phi(-2)$
- ☐ **E** none of them

Feedback at question level

$\Phi(1)$ is also considered correct, since indeed $\Phi(1) = 1 - \Phi(-1)$

Question 11 – random_experiment – 505564.4.0

Question type: Multiple response

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Final 2025

Answer option order: Random

Partial scoring: Yes

Maximum score: 1

Chance score: 0.33 pt. / 33%

Status:

Last modified: 09/10/2025 20:33

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Give all true statements about random experiments.

A If $P(\{\omega\}) = 0$ for every $\omega \in A$, then $P(A) = 0$.

B The sample space is the set of events.

C If $P(A) \neq 0$ and $P(B | A) \neq 0$, then $P(B) \neq 0$.

Question 12 – roll_dice – 505668.2.1

Question type: Fill in (numerical)

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Final 2025

Answer option order: Fixed

Partial scoring: No

Maximum score: 1

Chance score: 0.00 pt. / 0%

Status:

Last modified: 22/10/2025 17:50

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

A person rolls a fair dice twice. What is the probability that the result of the first roll is greater than 4 and, at the same time, the result of the second roll is less than the result of the first roll? Round to two decimals.

0.25

Feedback at question level

$P(A) = \frac{9}{36}$ with $A = \{(5,1),(5,2),(5,3),(5,4),(6,1),(6,2),(6,3),(6,4),(6,5)\}$

Question 13 – sample mean – 508710.1.2

Question type: Fill in (numerical)

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Final 2025

Answer option order: Fixed

Partial scoring: No

Maximum score: 1

Chance score: 0.00 pt. / 0%

Status:

Last modified: 22/10/2025 17:52

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Assume that a random experiment is performed 20 times and the outcome is modelled by random variables $X_1, \dots, X_{20}$ with expected value $\mu = 3$ and variance $\sigma^2 = 4$. What is the variance of the sample mean of $X_1, \dots, X_{20}$? Give one decimal.

0.2

Feedback at question level

= 4/20

Question 14 – sample variance – 508862.1.1

Question type: Fill in (numerical)

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Final 2025

Answer option order: Fixed

Partial scoring: No

Maximum score: 1

Chance score: 0.00 pt. / 0%

Status:

Last modified: 22/10/2025 17:54

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Consider a random variable X with unknown expected value μ and variance σ^2 . Assume that a realisation x_1, x_2, x_3 of the random sample $X_1, X_2, X_3 \sim X$ is given by $x_1 = -2, x_2 = 0, x_3 = 2$. Compute the corresponding sample variance S^2 .

4

Feedback at question level

= $\frac{1}{3-1} * ((-2)^2 + 0^2 + 2^2)$

Question 15 – variance – 508702.1.1

Question type: Multiple choice

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Final 2025

Answer option order: Random

Partial scoring: No

Maximum score: 1

Chance score: 0.25 pt. / 25%

Status:

Last modified: 22/10/2025 17:57

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

From a random variable X with range $\{-2, +2\}$ it is known that $P(X = -2) = P(X = +2)$. What is the variance of X ?

- A 2
- ☒ B 4
- C 16
- D none of the given values

Feedback at question level

$= (-2)^2 * P(X = -2) + (+2)^2 * P(X = +2)$ with $P(X = -2) = P(X = +2) = 1/2$

Question 16 – central limit theorem – 508711.1.2

Question type: Fill in (numerical)

Pre-test item: No

Folder: /Top/Personal/My Questions oft300/Statistical methods Final 2025

Answer option order: Fixed

Partial scoring: No

Maximum score: 1

Chance score: 0.00 pt. / 0%

Status:

Last modified: 22/10/2025 17:58

Attributes: *Taxonomie/Taxonomy* 00 Onbekend/Unknown

Let $X_1, \dots, X_{1000}$ be independent, identically distributed random variables with expectation value $\mu=0.4$ and variance $\sigma^2=1$. Then $P(X_1 + \dots + X_{1000} \leq 400)$ is approximately equal to which number? Give one decimal.

0.5

Feedback at question level

$= \Phi(0) = 1/2$

Test matrix

No matrix defined

